Lección 6

M. Siddikov

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Plan de trabajo

Formulación Lagrangiana

– Resumen de conceptos principales

... Principio de mínima acción y variables generalizadas

– Teorema de Noether: Simetrías→Integrales de movimiento

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana
2.1. Variables Generalizadas
2.2. Clasificación de los Sistemas Mecánicos
2.3. Elementos de Cálculo Variacional
2.4. Ecuaciones de Lagrange
2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

UNIDAD 3. Teoremas de Conservación
3.1. Teorema de Noether.
3.2. Aplicaciones del teorema de Noether.

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

FIS-426:

1) Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Hamiltonianos
- Otros principios variacionales
- Simetrías y Leyes de Conservación (Teorema de Noether)

Aviso

Tarea 1

- En Aula Virtual, Sección "Evaluaciones" del módulo 1 "Teoría de la Relatividad Especial"
- 2 problemas de desarrollo:
  - Cinemática relativista, variables de Mandelstam
  - Movimiento de partícula relativista en campo electromagnético
- Fecha de entrega: **08/04/2026**

Certamen 1 del año 2025 (con solución, solicitud de alumnos):

- En Aula Virtual, Sección "Actividades" del módulo 1 "Teoría de la Relatividad Especial"
- Materiales de Unidades I y II

Lagrangian formalism (summary)

- Formulation in terms of Equations of Newton is useful, yet is not unique.

- Lagrangian formulation is based on **Hamilton's minimal action principle:**
  - *For each dynamical system there is a functional called "action"* $S = \int_{t_1}^{t_2} L(q_i, \dot{q}_k, t)\, dt$ *such that real trajectories correspond to the minimum of the action. The function* L *is Lagrangian of the system.*
  - It is **alternative (≈equivalent) to Newtonian formulation for conservative forces:**
  - *If we choose* $L = T - U = m\dot{\boldsymbol{q}}^2/2 - U(\boldsymbol{q})$, *then 2nd law of Newton* $m\ddot{\boldsymbol{q}} = -\boldsymbol{\nabla} U(\boldsymbol{q})$ *is equivalent to* ***Euler-Lagrange equations***

$$0 = \frac{\delta \mathcal{F}}{\delta q_a} = \frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right), \quad a = 1, \ldots, 3, \quad p_a \equiv \frac{\partial L}{\partial \dot{q}_a}$$

Advantages of Lagrange formulation

–More convenient & easy in curvilinear coordinates, noninertial reference frame ...
–Easy to implement conservation laws
–Easy to implement constraints

Drawback of Lagrange formulation

–Applicable only for conservative forces ($\boldsymbol{F} = -\boldsymbol{\nabla} U$),
... not for dissipative forces (friction, ...).

Nonuniqueness of the lagrangian

The Lagrangian is not uniquely defined: multiplication of lagrangian by arbitrary **constant**,

$$L\left(q, \dot{q}, t\right) \to \lambda\, L\left(q, \dot{q}, t\right), \qquad \lambda = \mathrm{const}$$

essentially does NOT affect the equations of motion (all terms are multiplied by the same const λ which eventually vanishes):

$$0 = \lambda \left[\frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right)\right], \quad a = 1, \ldots, 3, \quad p_a \equiv \frac{\partial L}{\partial \dot{q}_a}$$

–Similarly, can check that adding a full derivative (d/dt) of *arbitrary* function $f(q, t)$

$$L \to L + \frac{d\, f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a}$$

does not change the equations of motion and the trajectory.

Proof 1:

Minimal action principle (Hamilton)

$$S = \int_{t_1}^{t_2} L\left(q_i,\ \dot{q}_k,\ t\right) dt = \min$$

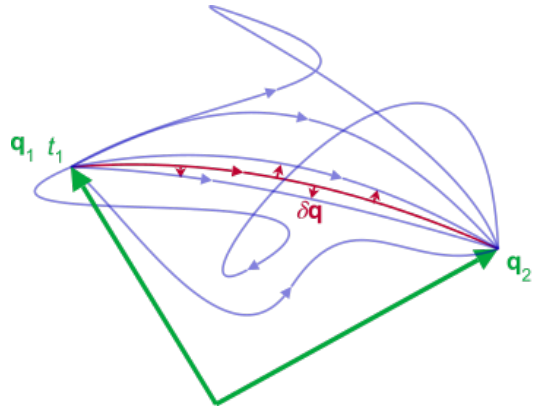
We can see that the transformation of Lagrangian

$$L \to L + \frac{d\, f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a},$$

leads to modification of action

$$S \to S + \int_{t_1}^{t_2} dt\, \frac{d\, f(q,t)}{dt} = S + f(q_2, t_2) - f(q_1, t_1)$$

-while S changes, this change only depends on the endpoint values of f. The value of S does not change when we vary the curve at intermediate point, and thus modification of lagrangian cannot affect equation of motion.



Proof 2:

Let's show explicitly that Euler-Lagrange equations remain invariant under

$$L \to L + \frac{d\,f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a}$$

Change of different contributions in Euler-Lagrange equation:

$$\frac{\partial L}{\partial q_a} \to \frac{\partial L}{\partial q_a} + \frac{\partial^2 f(q,t)}{\partial q_a\,\partial t} + \sum_b \dot{q}_b \frac{\partial^2 f(q,t)}{\partial q_a \partial q_b}\,,$$

$$\frac{\partial L}{\partial \dot{q}_a} \to \frac{\partial L}{\partial \dot{q}_a} + \frac{\partial f(q,t)}{\partial q_a}\,,$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) \to \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) + \frac{d}{dt}\left(\frac{\partial f(q,t)}{\partial q_a}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) + \frac{\partial^2 f(q,t)}{\partial q_a\,\partial t} + \sum_b \dot{q}_b \frac{\partial^2 f(q,t)}{\partial q_a \partial q_b}$$

So we can see that in the difference

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right),$$

all the red-colored terms will cancel, as expected.

Example: External EM field

Assume that You have a lagrangian

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} + e\,\dot{\boldsymbol{r}} \cdot \boldsymbol{A}(\boldsymbol{r}) - e\,\varphi(\boldsymbol{r})$$

where $\boldsymbol{A}, \varphi$ are some external fields.

– The "gauge transformation"

$$\varphi \to \varphi - \frac{\partial f}{\partial t}, \quad \boldsymbol{A} \to \boldsymbol{A} + \nabla f$$

where $f(\boldsymbol{r}, t)$ is arbitrary function, modifies the lagrangian as

$$L \to L + e\frac{\partial f}{\partial t} + e\dot{\boldsymbol{r}} \cdot \boldsymbol{\nabla} f = L + e\frac{df}{dt}$$

and thus does not change equations of motion

Lemma:

– This lagrangian corresponds to motion of a charged particle in external electromagnetic field.

$$\Rightarrow m\,\ddot{\boldsymbol{r}} = e\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right), \qquad \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \quad \boldsymbol{E} = -\boldsymbol{\nabla}\varphi - \frac{\partial \boldsymbol{A}(\boldsymbol{r})}{\partial t}$$

Example 2: External EM field (proof)

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} + e\,\dot{\boldsymbol{r}}\cdot\boldsymbol{A}(\boldsymbol{r}) - e\,\varphi(\boldsymbol{r})$$

Generalized momentum:

$$p_a = \frac{\partial L}{\partial \dot{r}_a} = m\,\dot{\boldsymbol{r}} + e\,\boldsymbol{A}(\boldsymbol{r}) \neq m\,\dot{\boldsymbol{r}}$$

Equations of motion:

$$\frac{d\,\boldsymbol{p}}{dt} = m\,\ddot{\boldsymbol{r}} + e\,\frac{d\boldsymbol{A}(\boldsymbol{r})}{dt} = -e\partial_a\varphi(\boldsymbol{r}) + e\,\boldsymbol{\nabla}_A\,(\boldsymbol{v}\cdot\boldsymbol{A})$$

$$\frac{d\boldsymbol{A}(\boldsymbol{r})}{dt} \equiv \frac{\partial\boldsymbol{A}(\boldsymbol{r})}{\partial t} + (\boldsymbol{v}\cdot\nabla)\,\boldsymbol{A}(\boldsymbol{r})$$

$$\boldsymbol{v}\times[\boldsymbol{\nabla}\times\boldsymbol{A}] = \nabla_A\,(\boldsymbol{v}\cdot\boldsymbol{A}(\boldsymbol{r})) - (\boldsymbol{v}\cdot\nabla)\,\boldsymbol{A}(\boldsymbol{r})$$

$$\Rightarrow m\,\ddot{\boldsymbol{r}} = e\,(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B})\,, \qquad \boldsymbol{B} = \boldsymbol{\nabla}\times\boldsymbol{A}, \quad \boldsymbol{E} = -\boldsymbol{\nabla}\varphi - \frac{\partial\boldsymbol{A}(\boldsymbol{r})}{\partial t}$$

Lagrangian & Galilean transformations

Assume that a system of particles is *closed*, *i.e.* particles interact only with each other, so the potential energy is given by $U = U(|\boldsymbol{q}_a - \boldsymbol{q}_b|)$. The Lagrangian is NOT invariant under Galilean transformations, however the Euler-Lagrange do not change their form (are "covariant").

- Change of velocities:

$$\boldsymbol{v} \to \boldsymbol{v} + \boldsymbol{V},$$

- Change of Lagrangian under Galilean transformation:

$$L \to L + \boldsymbol{V} \cdot \left(\sum_a m_a \boldsymbol{v}_a \right) + \left(\sum_a m_a \right) \frac{\boldsymbol{V}^2}{2} =$$

$$= L + \frac{d}{dt} \left[\boldsymbol{V} \cdot \left(\sum_a m_a \dot{\boldsymbol{q}}_a \right) + \left(\sum_a m_a \right) \frac{\boldsymbol{V}^2 t}{2} \right]$$

– The lagrangian is NOT invariant under Galilean transformations

– The additional term is a full derivative, so it will not affect equations of motion.

Example 2

Find the equation of motion for a system with lagrangian given by

$$L = e^{-\gamma t}\left(\frac{m\,\dot{\boldsymbol{x}}^2}{2} - \frac{k\boldsymbol{x}^2}{2}\right)$$

Example 2 (solution)

$$L = e^{\gamma t}\left(\frac{m\,\dot{\boldsymbol{x}}^2}{2} - \frac{k\boldsymbol{x}^2}{2}\right)$$

$$\frac{\partial L}{\partial \boldsymbol{x}} = -k\,\boldsymbol{x}e^{-\gamma t}, \quad \frac{\partial L}{\partial \dot{\boldsymbol{x}}} = e^{\gamma t} m\dot{\boldsymbol{x}}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{x}}}\right) = \frac{d}{dt}\left(e^{\gamma t} m\dot{\boldsymbol{x}}\right) = \left(m\ddot{\boldsymbol{x}} + \gamma m\,\dot{\boldsymbol{x}}\right)e^{\gamma t}$$

$$\left(m\ddot{\boldsymbol{x}} + \gamma m\,\dot{\boldsymbol{x}}\right)\cancel{e^{-\gamma t}} = -k\,\boldsymbol{x}\cancel{e^{-\gamma t}}$$

$$m\ddot{\boldsymbol{x}} = -\gamma m\,\dot{\boldsymbol{x}} - k\boldsymbol{x}$$

So we got EOM in presence of velocity-dependent absorptive force (instead of $k\boldsymbol{x}^2/2$ could have any $U(\boldsymbol{x})$)

– This is exceptional case: if we have any other velocity dependence in absorptive force, or kinetic energy $T \neq mv^2/2$, this method won't work.

Control questions

So far we considered lagrangians which include only $q(t)$ and $\dot{q}(t)$.

$-$Can the lagrangian include higher order derivatives, *e.g.* $\ddot{q}$, $\dddot{q}$?

$-$Can the lagrangian L include nonlocal terms, $e.g.$:

$$\dot{q}(t) \int d\tau \; K(t, \tau) \dot{q}(\tau)$$

where $K(t, \tau)$ is some kernel

Please explain in detail ...

Properties of Lagrangian

– For more than one particle generalization is

$$L = T - U = \sum_{a=1}^{3n} \frac{m\,\dot{q}_a^2}{2} - U\left(q_1, ..., q_{3n}\right) \qquad (2)$$

–Lagrangian (action) is defined only for the whole system, NOT for individual particles. It depends on all coordinates q_a and their derivatives $\dot{q}_a$.

–If the system includes 2 <u>non-interacting</u> parts, then its lagrangian is given by a sum

$$L = L_1 + L_2,$$

$$L_1 = \sum_{a=1}^{k} \frac{m\,\dot{q}_a^2}{2} - U_1\left(q_1, ..., q_k\right)$$

$$L_2 = \sum_{a=k+1}^{n} \frac{m\,\dot{q}_a^2}{2} - U_2\left(q_{k+1}, ..., q_n\right)$$

so the equations of motion (equations of Euler-Lagrange) for 2 subsystems decouple

Lagrangian formalism - curvilinear coordinates

- Earlier we saw that in Cartesian coordinates Euler-Lagrange equations coincide with Newtonian equations in Cartesian coordinates

$$m\ddot{\boldsymbol{q}} = -\boldsymbol{\nabla} U(\boldsymbol{q})$$

- For some problems (e.g. motion in spherically symmetric field) it may be more convenient to use other reference frame.

  ***Caution:** In curvilinear coordinates the basis vectors $\boldsymbol{e}_a \neq \mathrm{const}$

$$\frac{d\boldsymbol{V}}{dt} = \sum_a \frac{d\left(V_a \boldsymbol{e}_a\right)}{dt} = \sum_a \frac{dV_a}{dt}\boldsymbol{e}_a + \sum_a V_a \frac{d\boldsymbol{e}_a}{dt}$$

  where $\boldsymbol{e}_a$ are basis vectors at the particle's instant location, and

$$\frac{d\boldsymbol{e}_a}{dt} = \underbrace{\frac{\partial \boldsymbol{e}_a}{\partial t}}_{0} + \frac{\partial \boldsymbol{e}_a}{\partial q_b}\frac{\partial q_b}{\partial t}$$

  $\Rightarrow$ In Newtonian equations may get some "additional" terms (can mix components).

- Now we are going to analyze how the equations of Euler-Lagrange look in other reference frames

Lagrangian formalism - curvilinear coordinates

Assume that now we make transformation of variables (e.g. from Cartesian system to spherical or cylindrical),

$$q_a = q_a\left(r_1,\, r_2,\, r_3\right), \qquad a = 1, 2, 3,$$

Demonstrate that the Lagrangian in terms of new variables $\{r_a, \dot{r}_a\}$ is given by

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a \dot{r}_b}{2} - U\left(r_a\right), \quad \mu_{ab}(r) = m \sum_{k=1}^{3} \frac{\partial q_k}{\partial r_a} \frac{\partial q_k}{\partial r_b} = \mu\, g_{ab}$$

– Tensor g_{ab} is known as **metric tensor.** May be NOT diagonal in non-orthogonal coordinates

* In **orthogonal curvilinear coordinates** (spherical, cylindrical, elliptical, parabolic, ...)

$$g_{ab} = h_a^2 \delta_{ab}, \qquad \Rightarrow \mu_{ab} = \delta_{ab} \;\left(m\, h_a^2\right)$$

where h_a are called **Lamé coefficients,** may be found in any mathematical book or Wikipedia

$$\Downarrow$$

We don't need to evaluate μ_{ab} from scratch, may immediately write it out using known h_a.

Transformation of variables:

$$q_a = q_a\left(r_1,\, r_2,\, r_3\right),$$

The lagrangian in terms of variables r_i:

$$\dot{q}_a = \sum_{b=1}^{3} \frac{\partial q_a}{\partial r_b}\, \dot{r}_b$$

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\, \dot{r}_a \dot{r}_b}{2} - U\left(r_a\right), \quad \mu_{ab}(r) = m \sum_{k=1}^{3} \frac{\partial q_k}{\partial r_a} \frac{\partial q_k}{\partial r_b}$$

Relation to metric tensor g_{ab}:
−need to focus on kinetic energy transformation:

$$T_{\text{Cartesian}} = \frac{m}{2}\left(\frac{d\boldsymbol{r}}{dt}\right)^2 = \sum_{a,b=1}^{3} \underbrace{m\, g_{ab}}_{\mu_{ab}} \frac{dr_a}{dt} \frac{dr_b}{dt},$$

−we used that interval between two points by definition of g_{ab} is given by

$$(d\boldsymbol{r})^2 = \sum_{a,b=1}^{3} g_{ab} dr_a dr_b$$

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a \dot{r}_b}{2} - U\left(r_a\right), \quad \mu_{ab}(r) = m \sum_{k=1}^{3} \frac{\partial q_k}{\partial r_a} \frac{\partial q_k}{\partial r_b}$$

Example:

Transformation of variables (Cartesian→spherical):

$$q_x = r \sin\theta \cos\phi, \quad q_y = r \sin\theta \sin\phi, \quad q_z = r \cos\theta.$$

The lagrangian in terms of variables r, θ, ϕ.

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a \dot{r}_b}{2} - U\left(r_a\right)$$

$$T = m \frac{(\dot{r})^2 + r^2(\dot{\theta})^2 + r^2 \sin^2\theta\,(\dot{\phi})^2}{2}$$

Technical detail: explicit evaluation of μ_{ab}

Transformation of variables:

$$q_x = r\sin\theta\cos\phi,$$
$$q_y = r\sin\theta\sin\phi,$$
$$q_z = r\cos\theta.$$

$$\mu_{ab}(r) = m\sum_{k=1}^{3}\frac{\partial q_k}{\partial r_a}\frac{\partial q_k}{\partial r_b}$$

$$\begin{aligned}\frac{\mu_{rr}}{m} &= \frac{\partial x}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial z}{\partial r}\frac{\partial z}{\partial r}\\ &= \sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi\\ &+ \cos^2\theta =\\ &= \sin^2\theta\underbrace{\left(\cos^2\phi + \sin^2\phi\right)}_{=1} +\\ &+ \cos^2\theta = 1\end{aligned}$$

$$\begin{aligned}\frac{\mu_{r\theta}}{m} &= \frac{\partial x}{\partial r}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial \theta} + \frac{\partial z}{\partial r}\frac{\partial z}{\partial \theta}\\ &= \sin\theta\cos\phi\, r\,\cos\theta\cos\phi +\\ &+ \sin\theta\sin\phi\, r\,\cos\theta\sin\phi\\ &- r\sin\theta\cos\theta = \ldots = 0\end{aligned}$$

$$\begin{aligned}\frac{\mu_{\theta\theta}}{m} &= \frac{\partial x}{\partial \theta}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial \theta}\frac{\partial y}{\partial \theta} + \frac{\partial z}{\partial \theta}\frac{\partial z}{\partial \theta}\\ &= (r\,\cos\theta\cos\phi)^2 +\\ &+ (r\,\cos\theta\sin\phi)^2 +\\ &+ (r\,\sin\theta)^2 = r^2\end{aligned}$$

etc.

Explicit expression for μ_{ab} in any curvilinear coordinates

– **Cartesian**:

$$\mu_{ab}^{(\mathrm{Cart})} = m\delta_{ab}$$

– **Spherical**:

$$\mu_{ab}^{(\mathrm{Sph.})} = \begin{pmatrix} m & & \\ & mr^2 & \\ & & mr^2\sin^2\theta \end{pmatrix}$$

$$T = \frac{m\dot{r}^2 + mr^2\dot{\theta}^2 + mr^2\sin^2\theta\dot{\varphi}^2}{2}$$

– **Cylindrical:**

$$\mu_{ab}^{(\mathrm{Cyl.})} = \begin{pmatrix} m & & \\ & mr^2 & \\ & & m \end{pmatrix},$$

$$T = \frac{m\dot{r}^2 + mr^2\dot{\varphi}^2 + m\dot{z}^2}{2}$$

Lemma

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) \quad (1)$$

If we make transformation of coordinates

$$q_a = q_a\left(r_1,\, r_2,\, r_3\right),$$

then the Euler-Lagrange equations in terms of variables r_a have a form

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = 0, \quad (2)$$

i.e. the Euler-Lagrange equations are *covariant*.

Proof:

$$\frac{\partial L}{\partial q_k} = \sum_{a=1}^{3} \frac{\partial r_a}{\partial q_k} \frac{\partial L}{\partial r_a},$$

$$\frac{\partial L}{\partial \dot{q}_k} = \sum_{a=1}^{3} \frac{\partial r_a}{\partial q_k} \frac{\partial L}{\partial \dot{r}_a}, \qquad \dot{q}_k = \frac{\partial q_k}{\partial r_a} \dot{r}_a$$

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \sum_{a=1}^{3} \underbrace{\frac{\partial r_a}{\partial q_k}}_{\mathcal{M}_{ka}} \underbrace{\left[\frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right)\right]}_{X_a} = 0$$

If $\det(\mathcal{M}) = \det\left(\frac{\partial r_a}{\partial q_k}\right) \neq 0$, then equations

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) \qquad (1)$$

$$0 = \frac{\delta \mathcal{F}}{\delta r_a} = \frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = 0 \qquad (2)$$

are equivalent.

Lagrangian formalism

Covariance of Euler-Lagrange equations means that we can write out the Lagragian and derive Euler-Lagrange equations in *any* reference frame (not only Cartesian!). We do not need to write out forces, make their projections etc. This is one of the important properties of the approach which makes it so useful.

In what follows we'll use a term "***generalized coordinates***" for *any* set of variables $\{q_a\}$ which permit to characterize the position of all the particles in the system (without specifying if it is Cartesian, spherical, elliptical ...) We'll call "***generalized momenta***" and "***generalized forces***" the variables

$$p_k \equiv \frac{\partial L}{\partial \dot{q}_k}, \qquad f_k = \frac{\partial L}{\partial q_k}$$

so the Euler-Lagrange equation gets the form similar to 2nd law of Newton:

$$\dot{p}_k = f_k = \frac{\partial L}{\partial q_k}.$$

Generalized momenta - curvilinear coordinates

Definition: **"generalized momenta"** are the derivatives

$$p_k \equiv \frac{\partial L}{\partial \dot{q}_k}$$

Examples:

– In Cartesian coordinates for free particles $p_i = m\dot{q}_i$

– In spherical coordinates for the same free particles we have

$$L = T = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a \dot{r}_b}{2} = m\frac{(\dot{r})^2 + r^2(\dot{\theta})^2 + r^2 \sin^2\theta\,(\dot{\phi})^2}{2}$$

$$p_r = m\,\dot{r}, \qquad p_\theta = m\,r^2\dot{\theta}, \qquad p_\phi = m\,r^2 \sin^2\theta\dot{\phi}$$

$\Rightarrow$ p_i is not just "velocity" (derivative of coordinate), may include extra (coordinate-dependent) prefactors

Control question

$$L = T - U = \sum_{ab} \frac{\mu_{ab}(r)\, \dot{r}_a \dot{r}_b}{2} - U(r)$$

–For concreteness, You may assume that we have Coulomb potential, $U(r) \sim -\alpha/r$

- The generalized momenta in curvilinear coordinates:

$$p_a = \frac{\partial L}{\partial \dot{r}_a} = \sum_b \mu_{ab}(r)\, \dot{r}_b$$

The derivatives of the Lamé coefficients will give us contributions ("forces") which are bilinear in velocities,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = \sum_b \mu_{ab}(r)\, \ddot{r}_b + \sum_b \frac{\partial\, \mu_{ab}(r)}{\partial r_c}\, \dot{r}_b \dot{r}_c$$

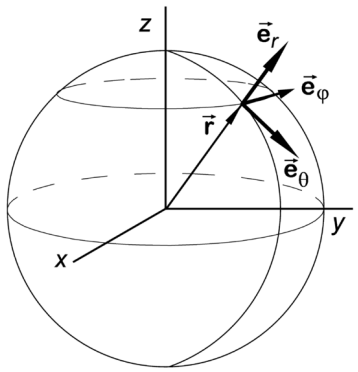
Does this result agree with the second law of Newton?
Please explain in detail ...

Control question (reply)

To understand what happens, we'll start from concrete example: Newtonian equations in spherical coordinates, and later will provide a general reply

Relation of spherical and Cartesian basis vectors

(use notation $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$, $\boldsymbol{e}_\varphi$ instead of $\hat{\mathrm{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\varphi}}$ to avoid confusion):



$$\boldsymbol{e}_r = \sin\theta\cos\varphi\,\boldsymbol{e}_x + \sin\theta\sin\varphi\,\boldsymbol{e}_y + \cos\theta\,\boldsymbol{e}_z$$
$$\boldsymbol{e}_\theta = \cos\theta\cos\varphi\,\boldsymbol{e}_x + \cos\theta\sin\varphi\,\boldsymbol{e}_y - \sin\theta\,\boldsymbol{e}_z$$
$$\boldsymbol{e}_\varphi = -\sin\varphi\,\boldsymbol{e}_x + \cos\varphi\,\boldsymbol{e}_y$$

$$\frac{d\boldsymbol{e}_r}{dt} = \dot{\theta}\boldsymbol{e}_\theta + \sin\theta\dot{\varphi}\boldsymbol{e}_\varphi$$
$$\frac{d\boldsymbol{e}_\theta}{dt} = -\dot{\theta}\boldsymbol{e}_r + \dot{\varphi}\cos\theta\boldsymbol{e}_\varphi$$
$$\frac{d\boldsymbol{e}_\varphi}{dt} = -\dot{\varphi}\left(\cos\varphi\,\boldsymbol{e}_x + \sin\varphi\,\boldsymbol{e}_y\right) = -\dot{\varphi}\left(\boldsymbol{e}_\theta\cos\theta + \boldsymbol{e}_r\sin\theta\right)$$

Vectors in sphercal coordinates

Radius-vector of particle in spherical coordinate, its velocity and acceleration:

$$\boldsymbol{r} = r\,\boldsymbol{e}_r,$$

$$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = \dot{r}\boldsymbol{e}_r + r\left(\dot{\theta}\boldsymbol{e}_\theta + \sin\theta\dot{\varphi}\boldsymbol{e}_\varphi\right),$$

$$\begin{aligned}\frac{d\boldsymbol{v}}{dt} &= \ddot{r}\boldsymbol{e}_r + \dot{r}\left(\dot{\theta}\boldsymbol{e}_\theta + \sin\theta\dot{\varphi}\boldsymbol{e}_\varphi\right) + r\left(\ddot{\theta}\boldsymbol{e}_\theta + \sin\theta\ddot{\varphi}\boldsymbol{e}_\varphi + \cos\theta\dot{\varphi}\dot{\theta}\boldsymbol{e}_\varphi\right) + \\ &+ \dot{r}\frac{d\boldsymbol{e}_r}{dt} + r\left(\dot{\theta}\frac{d\boldsymbol{e}_\theta}{dt} + \sin\theta\dot{\varphi}\frac{d\boldsymbol{e}_\varphi}{dt}\right) = \\ &= \boldsymbol{e}_r\left(\ddot{r} - r\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right)\right) + \boldsymbol{e}_\theta\left(r\,\ddot{\theta} + 2\,\dot{r}\dot{\theta} - \frac{r^2\sin 2\theta\,\dot{\varphi}^2}{2}\right) + \\ &+ \boldsymbol{e}_\varphi\left(r\,\sin\theta\ddot{\varphi} + 2r\cos\theta\dot{\varphi}\dot{\theta} + 2\dot{r}\sin\theta\dot{\varphi}\right)\end{aligned}$$

- Can clearly see that the terms in red stem from derivatives of the basis vectors ($\vec{\boldsymbol{e}}_r$, $\vec{\boldsymbol{e}}_\theta$, $\vec{\boldsymbol{e}}_\varphi$).

Motion in spherical coordinates: Newton vs Lagrange

- Second law of Newton:

$$\frac{d\vec{\boldsymbol{p}}}{dt} = \boldsymbol{e}_r m\left(\ddot{r} - r\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right)\right) + \boldsymbol{e}_\theta m\left(r\,\ddot{\theta} + 2\,\dot{r}\dot{\theta} - \frac{r^2\sin 2\theta\,\dot{\varphi}^2}{2}\right) +$$

$$+\,\boldsymbol{e}_\varphi m\left(r\,\sin\theta\ddot{\varphi} + 2r\cos\theta\dot{\varphi}\dot{\theta} + 2\dot{r}\sin\theta\dot{\varphi}\right) = -\nabla U = -\frac{\partial U}{\partial r}\boldsymbol{e}_r \qquad (1)$$

- Explicit Euler-Lagrange equations:

$$m\ddot{r} \equiv m\frac{d^2 r}{dt^2} = -\frac{\partial U}{\partial r} + mr\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right), \qquad (2a)$$

$$\frac{d(m\,r^2\dot{\theta})}{dt} \equiv m\,r^2\ddot{\theta} + 2mr\,\dot{r}\dot{\theta} = \frac{m\,r^2\sin 2\theta\,\dot{\varphi}^2}{2} \qquad (2b)$$

$$\frac{d(m\,r^2\sin^2\theta\dot{\varphi})}{dt} = m\,r^2\sin^2\theta\ddot{\varphi} + m\,r^2\sin 2\theta\,\dot{\theta}\dot{\varphi} + 2m\,r\,\dot{r}\dot{\varphi}\sin^2\theta = 0 \qquad (2c)$$

- The equation (1) in components coincides with (2a-2c) up to common factors:

– should divide (2b) by r to get θ-component of (1)

– should divide (2c) by $r\sin\theta$ to get φ-component of (1)

$\Rightarrow$We got correct equations in Euler-Lagrange approach

–no need to analyze transformation of basis vectors ...

Motion in curvilinear coordinates: Newton vs Lagrange

- Our finding is NOT a coincidence
- Valid in any other curvilinear coordinates (cylindrical, elliptical, parabolic, ...)

  ... using $\boldsymbol{v} = \sum h_a \dot{r}_a \boldsymbol{e}_a$ we may find acceleration

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \sum h_a \ddot{r}_a \boldsymbol{e}_a + \sum \frac{\partial h_a}{\partial r^b} \dot{r}_a \dot{r}_b \boldsymbol{e}_a + \sum h_a \dot{r}_a \frac{d\boldsymbol{e}_a}{dt} \tag{1}$$

- **Courses of differential geometry:**

  "Derivatives" of basis vectors $d\boldsymbol{e}_a/dt$ may be related to Lame parameters ("metrics")

$$\frac{d\boldsymbol{e}_a}{dt} = \sum_b \frac{d\boldsymbol{e}_a}{dr_b} \dot{r}_b = \sum_{b,c} \Gamma^c_{ab} \boldsymbol{e}_c \dot{r}_b, \qquad \Gamma^c_{ab} = \left(\boldsymbol{e}^c \cdot \frac{d\boldsymbol{e}_a}{dr_b} \right) \tag{2}$$

  * (Christoffel) Symbols Γ^c_{ab} can be expressed via Lame parameters ("metrics") and their derivatives,

$$\Gamma^c_{ab} = \frac{1}{2} g^{cd} \left(\frac{\partial g_{ad}}{\partial r^b} + \frac{\partial g_{bd}}{\partial r^a} - \frac{\partial g_{ab}}{\partial r^d} \right), \qquad \mu_{ab} = m\, g_{ab}$$

  In orthogonal coordinates $g_{ab} = h_a^2 \delta_{ab}$, so Γ^c_{ab} will depend on h_a and its derivatives

$$\Gamma^c_{ab} = \left(\delta_{ac} \frac{\partial \ln h_a}{\partial x^b} + \delta_{bc} \frac{\partial \ln h_c}{\partial r^a} - \frac{\delta_{ab}}{2h_c^2} \frac{\partial h_a^2}{\partial r^c} \right), \tag{3}$$

  Substituting (2,3) into (1), it is possible to show that terms $\sim \dot{r}_a \dot{r}_b$ are the same in Euler-Lagrange equation and Newtonian 2nd law

Motion in curvilinear coordinates: Newton vs Lagrange

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Integrals of motion and Noether's theorem

Warm-up questions ...

Please formulate explicitly the Noether's theorem. Pay attention what conditions are needed for its validity
−Provide examples ...

Question 1

Why do we need the integrals of motion ? Should we consider them as additional condition on classical trajectory ?

Question 2

Assume that the action is invariant when we apply some continuous transformation. What exactly should we do in order to find the invariant Q?

Question 3

Assume that we have a continuous symmetry, which changes action as $S \to \lambda^n S$ but doesn't change the equation of motion. Are there any invariants associated with this symmetry?

Question 4

Assume that we know that $Q(q, \dot{q})$ is invariant on any classical trajectory. Is it possible to reconstruct the continuous symmetry, which corresponds to this invariant?

Integrals of motion & Noether's theorem

- EL equations are **the second order ODEs**, might be difficult to integrate directly
- Some systems of ODEs possess first integrals:
  - After some manipulations, ODEs can be reducible to a form

$$\frac{d}{dt} Q\left(q_a,\, \dot{q}_b,\, t\right) = 0 \qquad (1)$$

    so clearly along the classical trajectory

$$Q\left(q_a,\, \dot{q}_b, t\right) = \mathrm{const} \qquad (2)$$

    where the value of const may be fixed from initial condition
  - Since (2) is the 1st order ODE, Mathematically it is much simpler than 2nd order Euler-Lagrange equation
  - Existence of the integrals of motion is closely related to symmetries of the system

Noether's theorem

For any continuous symmetry group of the action $\mathcal{S}$, there is a conserved quantity (integral of motion).

- Symmetry group of the action=set of transformations which do not change S (might be translations, rotations, ...).
  - *Symmetry group must be continuous. If transformation is discrete (NOT continuous), Noether's theorem is not applicable.
- The Noether's theorem is valid in all branches of physics, always continuous symmetry corresponds to some integral of motion. We'll prove it only for classical mechanics below, but it might be extended (generalized) to other systems.

Integrals of motion & Noether's theorem

Noether's theorem

For any continuous symmetry of the action $\mathcal{S}$, there is a conserved quantity (integral of motion).

Examples:

- Invariance of L with respect to translations along $\boldsymbol{n}$

$$\boldsymbol{q} \to \boldsymbol{q} + \lambda\, \boldsymbol{n}, \quad \forall \lambda$$

  $\Rightarrow$conservation of momentum component $p_n = \boldsymbol{p} \cdot \boldsymbol{n}$

- Invariance of L with respect to rotations around $\hat{n}$ $\Rightarrow$conservation of angular momentum projection

$$M \cdot n \equiv n \cdot [\boldsymbol{q} \times \boldsymbol{p}]$$

- Time independence of L $\Rightarrow$conservation of energy

$$E \equiv \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} - L = T + U = \text{const}$$

Sometimes symmetries might be more complicated (e.g. potential $U(x - Vt)$ has symmetry with respect to $t \to t + t_0$, $x \to x + V\, t_0$). We'll learn a bit later how to construct invariants in this case.

Noether's theorem (proof)

Proof:

- Assume that $\epsilon = \mathrm{const}$ is a parameter of continuous symmetry transformation (translation, rotation, ...) which leaves the action S invariant
- Let's replace now $\epsilon = \mathrm{const}$ with $\epsilon = \epsilon(t) = \epsilon_0 + \delta\epsilon(t)$ with $\delta\epsilon\left(t_{a,b}\right) = 0$,. and $\delta\epsilon \ll \epsilon_0$
- First, let's assume that the transformation does not affect time, but changes the coordinates and velocities as:

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t), \qquad \dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t) + \dot{\epsilon}\gamma_a(t).$$

The classical trajectory is (local) minimum of the action, so small variations should NOT change S, since $\delta S/\delta q|_{q=q_{\mathrm{cl}}} = 0$. So we conclude that on classical trajectory $\delta S/\delta\epsilon(t) = 0$:

$$\delta S = \int_{t_1}^{t_2} dt\, \delta\epsilon \left(\frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}} \right) \right) = 0 \qquad (1)$$

Noether's theorem (II)

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t), \qquad \dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t) + \dot{\epsilon}\gamma_a(t).$$

The classical trajectory is (local) minimum of the action, so small variations should NOT change S, since $\delta S/\delta q|_{q=q_{\rm cl}} = 0$. So we conclude that on classical trajectory $\delta S/\delta\epsilon(t) = 0$:

$$\delta S = \int_{t_1}^{t_2} dt\, \delta\epsilon \left(\frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}}\right)\right) = 0 \qquad (1)$$

–The action S is invariant for transformations with $\epsilon = \mathrm{const}$, for this reason the lagrangian should be also invariant,

$$\frac{\partial L}{\partial \epsilon} \equiv \sum_a \left(\frac{\partial L}{\partial q_a}\gamma_a + \frac{\partial L}{\partial \dot{q}_a}\gamma_a \right) = 0 \qquad (2)$$

$$(1,2) \Rightarrow \frac{\partial L}{\partial \dot{\epsilon}} \equiv \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = Q = \mathrm{const}$$

▷We obtained explicitly the Noether invariant Q.

Noether's theorem: applications

Noether's theorem

Let's assume that the lagrangian of the system L with n degrees of freedom is invariant with respect to

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t),$$

$$\dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t),$$

$$t \to t$$

where $\epsilon = \mathrm{const}$ is a continuous parameter. Then the quantity

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a$$

is conserved.

Applications (examples):

►If the action's symmetry group corresponds to translations along the axis $\boldsymbol{n}$, then clearly $\gamma_a \equiv n_a$, so the associated integral of motion is

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = p \cdot n$$

-projection of momentum on $\boldsymbol{n}$

►If the action's symmetry group corresponds to rotations around $\boldsymbol{n}$, then

$$\delta\boldsymbol{q} = \delta\phi\, \boldsymbol{n} \times \boldsymbol{q} \Rightarrow \boldsymbol{\gamma} = \boldsymbol{n} \times \boldsymbol{q}$$

so the associated integral of motion is

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = \boldsymbol{p} \cdot [\boldsymbol{n} \times \boldsymbol{q}] =$$

$$= \boldsymbol{n} \cdot [\boldsymbol{q} \times \boldsymbol{p}] = \boldsymbol{n} \cdot \boldsymbol{M}$$

-projection of angular momentum $\boldsymbol{M}$ on the direction of $\boldsymbol{n}$

Noether theorem and symmetries

Noether's theorem

Let's assume that the lagrangian of the system L with n degrees of freedom is invariant with respect to

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t),$$

$$\dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t),$$

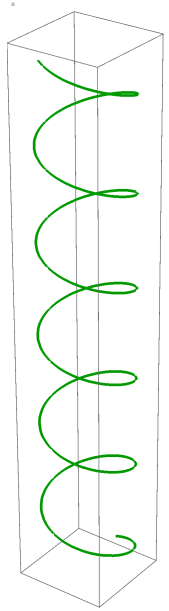
$$t \to t$$

where ϵ is a continuous parameter. Then the quantity

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a$$

is conserved.

Another example:



In cylindrical coordinates potential $U = U(r, z - a\phi)$ has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ \phi \to \phi + \frac{\epsilon}{a} \quad (1)$$

▷Since kinetic energy is invariant w.r.t. rotations and translations, transformation (1) is symmetry group of the action

▷ Components of the γ-vector:

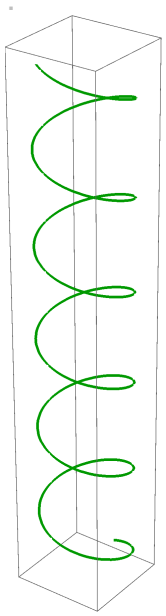
$$\gamma_r = 0,\ \gamma_\phi = \frac{\partial\phi}{\partial\epsilon} = a^{-1},\ \gamma_z = \frac{\partial z}{\partial\epsilon} = 1,$$

so the conserving quantity is

$$Q = \underbrace{\frac{\partial L}{\partial\dot{\phi}}}_{p_\phi \equiv M_z}\gamma_\phi + \underbrace{\frac{\partial L}{\partial\dot{z}}}_{p_z}\gamma_z = \frac{M_z}{a} + p_z$$

-neither p_z nor M_z conserve, yet their linear combination is constant (!)

Beware: only symmetries of S matter



▷ Action of a particle in external field:

$$S = \int dt\, L = \int dt \left[\frac{mv^2}{2} - U \right]$$

$$v^2 = g_{ij} v^i v^j$$

Compare 2 problems:

Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(r, z - a\phi)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ \phi \to \phi + \frac{\epsilon}{a} \quad (1)$$

Metrics in cylindr. coords: $g_{ij} = \{1, r^2, 1\}$ is invariant w.r.t. (1)
⇒There is integral of motion (Noether invariant) associated with symmetry (1) (which might be found taking derivative $\partial L / \partial \dot{\epsilon}$)

Assume that a charged particle moves in the external potential, which in cylindrical coordinates is given by $U = U(z - a\,r)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ r \to r + \frac{\epsilon}{a} \quad (2)$$

Metrics $g_{ij} = \{1, r^2, 1\}$ is NOT invariant w.r.t. (2)
⇒There is NO integrals of motion (Noether invariants) associated with symmetry (2)

- Our previous derivation was based on the fact that symmetry transformation does not affect (change) time. Now we'll try to see example when time is also changed. We'll start for simplicity with the simplest symmetry transformation

$$q_a(t) \to q_a(t), \quad \dot{q}_a(t) \to \dot{q}_a(t), \quad t \to t + \epsilon \qquad (1)$$

Noether's theorem

If the lagrangian L of the system is invariant with respect to transformation (1), where ϵ is a continuous parameter, then there is a conserved quantity (energy)

$$E = \sum_a \frac{\partial L}{\partial \dot{q}_a} \dot{q}_a - L = \mathrm{const}$$

Elementary proof:

$$\frac{dL}{dt} = \frac{\partial L}{\partial t} + \sum_a \left(\frac{\partial L}{\partial q_a} \dot{q}_a + \frac{\partial L}{\partial \dot{q}_a} \ddot{q}_a \right) = \frac{\partial L}{\partial t} + \frac{d}{dt} \sum_a \left(\frac{\partial L}{\partial \dot{q}_a} \dot{q}_a \right)$$

$$\Rightarrow \frac{dE}{dt} = \frac{d}{dt} \sum_a \left(\frac{\partial L}{\partial \dot{q}_a} \dot{q}_a - L \right) = -\frac{\partial L}{\partial t}$$

Since by definition S is invariant under transformation (1), i.e. L does not depend explicitly on time, $\partial L / \partial t = 0$ and thus $E = \mathrm{const}$.

Alternative proof:

- Let's replace now $\epsilon = \mathrm{const}$ with t-dependent $\epsilon = \epsilon(t) = \epsilon_0 + \delta\epsilon(t)$ with $\delta\epsilon\left(t_{a,b}\right) = 0_{,}$. and $\delta\epsilon \ll \epsilon_0$

- The transformation does not affect coordinates, but changes the time and velocities as:

$$t \to t + \epsilon(t), \quad dt \to dt\left(1 + \dot{\epsilon}(t)\right), \quad \dot{q}_a \equiv \frac{dq}{dt} \to \frac{\dot{q}_a}{1 + \dot{\epsilon}(t)} \approx \dot{q}_a\left(1 - \dot{\epsilon}(t)\right) + \mathcal{O}\left(\epsilon^2\right)$$

Change of action under this transformation:

$$S \equiv \int dt\, L(q, \dot{q}) \to \int dt\, \left(1 + \dot{\epsilon}(t)\right) L(q, \dot{q}(1 - \dot{\epsilon})) \approx \tag{1}$$

$$\approx \int dt\, L(q, \dot{q}) + \int dt\, \dot{\epsilon}\left(L(q, \dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a}\right) + \mathcal{O}\left(\epsilon^2\right)$$

$$S \approx \int dt\, L(q,\dot{q}) + \int dt\, \dot{\epsilon} \left(L(q,\dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} \right) + \mathcal{O}\left(\epsilon^2\right)$$

- As in previous derivation, we consider that on classical trajectory the variational derivative should vanish,

$$\left.\frac{\delta S}{\delta q}\right|_{q=q_{\mathrm{cl}}} = \frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}}\right) = 0 \qquad (2)$$

and $\partial L/\partial \epsilon = 0$ since for $\epsilon = \mathrm{const}$ the action S is invariant.

$$(1,2) \Rightarrow \frac{\partial L}{\partial \dot{\epsilon}} = L(q,\dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} \equiv -E = \mathrm{const} \qquad \mathrm{Q.E.D}$$

Now can generalize this result for arbitrary symmetry transformation which mixes coordinates and time:

$$q \to q_\epsilon, \quad t \to t_\epsilon, \qquad \epsilon \ll 1$$

1) Make Taylor expansion of S, picking up the linear $\mathcal{O}(\epsilon)$, $\mathcal{O}(\dot{\epsilon})$ terms
2) All the $\sim \mathcal{O}(\epsilon)$ terms will vanish since for $\epsilon = \mathrm{const}$ the action is invariant
3) All the linear terms $\sim \mathcal{O}(\dot{\epsilon})$ allow to find the desired Noether invariant

$$Q = \lim_{\varepsilon \to 0} \frac{\delta S\left(q_\epsilon, \dot{q}_\epsilon, t_\epsilon\right)}{\delta \dot{\epsilon}} = \mathrm{const}$$

4) General result for invariant Q:

$$S_\epsilon = \int dt_\epsilon \, L\left(q_\epsilon, \dot{q}_\epsilon, t\right) = \int dt \left(\frac{\partial t_\epsilon}{\partial t}\right) L\left(q_\epsilon, \dot{q}_\epsilon, t\right)$$

$$\Rightarrow Q = \lim_{\varepsilon \to 0} \frac{\partial L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right)}{\partial \dot{\epsilon}}, \quad L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right) = \left(\frac{\partial t_\epsilon}{\partial t}\right) L\left(q_\epsilon, \dot{q}_\epsilon, t\right)$$

Example

Assume that the particle's lagrangian is given by

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} - U(x, y, z - V\,t)$$

Identify symmetry group and construct the integral of motion. Explain your result.

- Symmetry transform (Cartesian):

$$z \to z + V\,\epsilon,\; t \to t + \epsilon, \qquad \epsilon = \mathrm{const}$$

- Consider $\epsilon = \epsilon(t)$

$$dt \to dt\,(1 + \dot{\epsilon}(t))$$

$$\dot{\boldsymbol{r}} \to \frac{\dot{\boldsymbol{r}} + V\,\dot{\epsilon}\hat{\boldsymbol{z}}}{1 + \dot{\epsilon}} \approx \dot{\boldsymbol{r}}(1 - \dot{\epsilon}) + V\,\dot{\epsilon}\hat{\boldsymbol{z}} + \mathcal{O}\left(\dot{\epsilon}^2\right)$$

Change of lagrangian

$$L \to L_\epsilon = L\,(1 + \dot{\epsilon}) - \dot{\epsilon}\dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\dot{\epsilon}\hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot{z}} + \mathcal{O}\left(\dot{\epsilon}^2\right)$$

Example

$$L \to L_\epsilon = L\,(1+\dot\epsilon) - \dot\epsilon \dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\dot\epsilon \hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot z} + \mathcal{O}\left(\dot\epsilon^2\right)$$

– Construct the Noether's invariant:

$$Q = \left.\frac{\partial L_\epsilon}{\partial \dot\epsilon}\right|_{\varepsilon\to 0} = L - \dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot z} = -E + p_z(t)V \quad (1)$$

Recall Galileoss transform for velocities and transformation of kinetic energy:

$$\frac{m\left(\boldsymbol{v}(t)-\boldsymbol{V}\right)^2}{2} = \frac{m\boldsymbol{v}^2(t)}{2} - \boldsymbol{p}(t)\cdot\boldsymbol{V} + \frac{mV^2}{2}$$

$V = \mathrm{const},\ v = v(t) = \dot r(t) \neq \mathrm{const}$

- Can rewrite (1) as

$$\frac{m\left(\boldsymbol{v}(t)-\boldsymbol{V}\right)^2}{2} = E_{\mathrm{rf}} + \frac{mV^2}{2}$$

where E_{rf} is the energy in the frame moving with velocity $-V\,\hat z$. In that frame

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} - U(x,y,z) + \frac{d}{dt}(\ldots)$$

does not depend on time, so energy is conserved

▷in contrast, in our frame potential depends on time, so $E \neq \mathrm{const}$.